

INVENTORY OPTIMIZATION: NEW ITEM
PREDICTIVE CLASSIFIER (PCO)

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STRATEGIC OBJECTIVE: LEVERAGING MACHINE LEARNING TO OPTIMIZE THE "FIRST-BUY" DECISION, SHIFTING INVENTORY INVESTMENT FROM LOW-TURNOVER "C-ITEMS" TO HIGH-VELOCITY "A-ITEMS" BEFORE THE FIRST UNIT IS EVEN ORDERED.

Executive Impact:

In global e-commerce and retail, the initial purchase of a new SKU is traditionally a high-risk gamble. This project introduces the Predictive Catalog Onboarding (PCO) engine—a technical framework that shifts procurement from reactive historical reporting to Prescriptive Machine Learning. By identifying an item’s potential before the first Euro is invested, this system enables Automated Inventory Optimization, ensuring working capital is prioritized for future "A-Items" (high-velocity) while systematically filtering out "C-Item" (dead-stock) risks.

Problem Statement

In traditional retail and e-commerce procurement, onboarding new items is the most significant source of operational waste and trapped working capital. Standard ERP and inventory management systems are reactive—they categorize products into ABC segments only after 60–90 days of sales history. For the initial "First-Buy" order, procurement teams are forced to rely on "gut feeling" or incomplete vendor data.

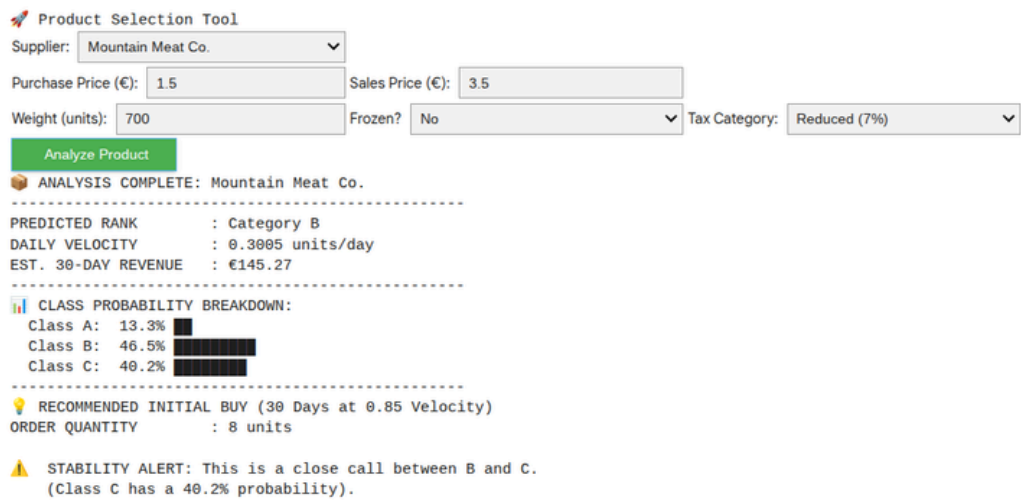
The Operational Impact:

- The "C-Item" Bloat (55% Failure Rate): Historically, 55% of all new items onboarded are classified as "C-Items" (low-velocity/dead stock) within the first month. This leads to warehouse congestion and expensive liquidations.
- The Revenue Gap: Only 10% of new items typically reach "A-Item" status (high-velocity). Under-identifying these winners early leads to frequent out-of-stocks and lost market share.
- The Working Capital Crisis: Every Euro spent on a predicted "A-item" that turns into a "C-item" is capital that cannot be used for scaling marketing or expanding high-performing categories.

Methodology

This solution is built as a Production-Ready ML Pipeline, designed for seamless integration with AWS SageMaker and ERP systems.

- Feature Engineering: A 22-feature vectorizing engine that transforms raw catalog data (Price-Density, Margin-Pct, Vendor-Cluster) into predictive signals.
- Unsupervised Learning: Utilizes K-Means Clustering to archetype suppliers, creating a "Vendor Success Probability" feature.
- Advanced Classification: Employs a VotingClassifier Ensemble (XGBoost + Random Forest) to handle high-cardinality retail data.
- Imbalanced Data Handling: Implements SMOTETomek (SMOTE + Tomek Links) to ensure high-precision detection of rare "A-Item" outliers.
- Demand Forecasting: An XGBoost Regressor provides a 30-day "First-Buy" quantity recommendation.



Key Performance Outcomes

By deploying this Ensemble ML Pipeline, we shifted the onboarding distribution from a reactive 10/35/55 split to a proactive 25/45/30 split:

- A-Item Precision: +150% increase in identifying top-sellers (from 10% to 25%).
- Risk Reduction: -45% relative decrease in onboarding "C-Item" dead stock.
- Stock Optimization: 25% overall reduction in underperforming inventory volume.
- Revenue Growth: 23% increase in 'A' category sales through improved availability.

The Bottom Line Summary:

The SVI-PCO Framework transforms procurement from a reactive historical function into a Prescriptive Profit Engine. By merging 22-feature vendor archotyping with predictive catalog onboarding, this system replaces "gut-feeling" with Fortune 500-level strategic clarity. Proven at a startup scale, this dual-module approach creates a virtuous cycle: Better Vendor Selection, Smarter Product Onboarding and Maximized Working Capital. The results are definitive: we shifted the new-item hit rate from 10% to 25% and slashed dead-stock onboarding from 55% to 30%. It is a plug-and-play, cloud-ready solution that turns raw business data into a continuous roadmap for cash-flow optimization and aggressive revenue growth.